

Why intelligence wins for OTM covered calls

A live session walkthrough of the SAGE signal engine — three stocks, four hours, and the cost of getting the gates wrong.

The setup

On the morning of June 2, three positions — CAT, CEG, and AAPL — opened higher with no existing covered calls. A simple rules-based system would have written calls at market open. SAGE instead evaluated each stock against a multi-gate momentum framework before placing any order.

The question this session answers: does the added complexity pay off, or does it just add friction and missed premium?

Why intelligence wins here

A simple rule — "if stock is up 0.5%, write a call" — would have fired at 9:05 on all three positions. At that point the move hadn't confirmed, stocks were still accelerating, and there was no pullback cushion. The calls would have been written at lower strikes with less premium, and all three stocks kept running anyway — pushing the positions immediately underwater.

The SAGE gates added friction, but the friction was informative. Each suppression logged the reason. Each config change was driven by live signal data. The system identified that the deceleration gate was generating false positives, that trend regime filtering needed broader defaults, and that ask-first pricing recovers meaningful premium over mid-price placement.

More importantly, every entry condition is captured in `scoring_factor_values` — IV rank, RSI, change%, pullback from high, Bollinger position, VIX, time of day. When these positions close, DANI will correlate those entry conditions to outcomes. Over hundreds of trades, that feedback loop is what separates a learning system from a spreadsheet macro.

The momentum gates — configured at session start

Gate	Setting	Status	Effect today
Min time ET	9:45 AM	Active	Blocked all signals until market settled
Min change %	0.5%	Active	CEG faded to -0.47% and never recovered
Gap-up filter	2% → 3%	Raised mid-session	CAT at +2.63% was blocked until raised

Deceleration gate	10 min lookback	Disabled mid-session	Noise — blocked CAT/AAPL on 0.06% drift
Pullback from high	0.17% → 0.12%	Lowered mid-session	AAPL held correctly near intraday high
Trend regime	Bullish/neutral	Added bearish	CAT was bearish — correctly identified
Max RSI	75	Active	CAT RSI 49.5 — passed cleanly

Session timeline

9:05 ET	Time gate blocks all signals CAT, CEG, AAPL logging every 5 min but suppressed. Simple system fires here — wrong.
9:45 ET	CEG suggestion fires — \$277.5 Jun 12, est \$485 premium Auto-STO just past time gate. CEG's one clean window. By 9:50 it was accelerating and blocked, then faded to -0.47%.
9:50 ET	Deceleration gate kills CAT and AAPL CAT: +2.53% to +2.59% (0.06% noise). AAPL: +0.85% to +0.92%. Both flagged accelerating.
10:20 ET	Config updated: deceleration off, bearish trend allowed, gap-up raised to 3% Each change driven by signal log data showing exactly which gate was firing and why.
10:43 ET	CAT auto-STO placed — \$915 Call Jun 5, 3 DTE, 1.9% OTM Placed at ask (\$7.18). Chase auto-started with mid as floor. Stock at \$888.
10:47 ET	CEG auto-STO placed — \$275 Call Jun 12, 10 DTE, 2.8% OTM Filled at \$5.10 limit exactly. ETrade routing worked cleanly.
Ongoing	AAPL climbing — now +2.0%, pullback gate correctly holding At \$312.52 and making new highs. Gate preventing a write at the absolute top.

Trades placed

Stock	Strike	Expiry	DTE	OTM%	Account	Limit	Status
CAT	\$915 Call	Jun 5	3	1.9%	Schwab 3866	\$7.18	Filled
CEG	\$275 Call	Jun 12	10	2.8%	ETrade 6917	\$5.10	Filled
AAPL	~\$318 Call	Jun 5-8	3-6	~2.9%	ETrade 6917	TBD	Pending

What this session taught SAGE

- ✗ **Deceleration gate is too noisy.** Comparing change% 10 minutes apart at hundredths of a percent generates false positives on flat, grinding moves. CAT at +2.53% to +2.59% is consolidation, not acceleration. Disabled permanently.

- ✓ **Pullback-from-high gate works as designed.** It correctly held AAPL during a genuine run-up, preventing a write at the absolute intraday peak where upside is maximally capped. The 0.12% threshold is a reasonable default.

- ✓ **Trend regime filtering is necessary — defaults need broadening.** CAT's bearish regime correctly reflected market structure. The filter should include bearish by default — writing covered calls into a down-trending stock on an up day is often the optimal entry.

- **Place at ask, chase to mid.** Prior logic placed at mid — both CAT and CEG showed -25% immediately after fill because market makers filled at bid anyway. New logic: start at ask, auto-chase down in \$0.05 steps with mid as the floor.

- **DANI needs outcome linkage to complete the loop.** Entry conditions are richly captured (IV rank 100, RSI 49.5, change +3.8%, pullback 0.04%, Bollinger %B 0.18). The missing wire: signal_id to final contract P&L; when the position closes.

AAPL position pending — case study will be updated with final entry details upon signal fire. All data sourced live from Schwab and ETrade APIs via the PRI Options Tracker.